R22

13/05/2026 SNK MED

Code No.: 22MEC36

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (Mech, CSE) VI Sem (Main/Backlog) Examination May 2026

Fundamentals of Design Thinking

Time: 3 Hours | Max Marks: 60

Note: "Answer ALL questions from Part-A and Part-B (including internal choices) at one place and in the same order as in the question paper."

Part - A

(5Q × 2M = 10 Marks)

			M	CO	BT
1		Give two examples of applications of design thinking in any engineered products/ services.	(2)	1	2
2		How is empathy different from sympathy?	(2)	2	2
3		What is the role of a user persona in design thinking process?	(2)	3	2
4		Mention at least four tools or techniques used in the ideation phase.	(2)	4	1
5		What is the significance of the last phase of design thinking? Give examples.	(2)	5	2

Part - B

(5Q × 10M = 50 Marks)

			M	CO	BT
6		Are design thinking and engineering activity interconnected or are they independent activities? Is it better to start with one and proceed to another sequentially? If yes, which one must come first and why? If not, justify their correlation or the lack of it. Use examples suitably.	(10)	1	3
		(OR)			
7	(a)	Write brief notes about linear thinking and lateral thinking with one example each.	(5)	1	1
	(b)	Represent the five stages of design thinking method and briefly mention the role of each stage.	(5)	1	1
8		Explain the importance of empathy for design thinking with suitable examples, particularly focusing on how immersive or experiential learning can help empathisation.	(10)	2	2
		(OR)			
9	(a)	Do you agree that empathisation is both necessary and sufficient to understand user needs? Justify your answer in any case.	(5)	2	4
	(b)	What is a beginner's mindset? Why is it important to get into such a mindset for user empathisation?	(5)	2	2
10		Consider a product or a service. Walk through the Define phase of design thinking for the chosen product/ service and explain the various steps to consider. Elaborate on suitable tools that may help the process.	(10)	3	2
		(OR)			
11	(a)	What is an ideal problem definition? How do you know that you have defined a problem properly?	(5)	3	2
	(b)	Write a brief note on empathy mapping with a suitable example explaining various aspects.	(5)	3	2

Explain with suitable examples how the Six Thinking Hats method helps ideation in design thinking. (10)

(OR)

(a) Explain the importance of Mind Maps used in design thinking. (5)

(b) How can brainstorming be used for effective ideation? (5)

Consider an existing product or a service. Explain how the value proposition is communicated through advertisements of the product/ service and outline specific aspects of user feedback that may have helped improve the product/ service. (10)

(OR)

(a) Explain the importance of Testing phase in design thinking. (5)

(b) What is the importance of iteration in the last phase of design thinking? Give a real-world example of a product/ service, explain how iteration can help or how the lack of iteration can be a problem. (5)

